

1. Which of the following is the unit for frequency of a vibration?

- (A) m
- (B) s
- (C) s^{-1}
- (D) $m s^{-1}$

2. Which of the following pairs consist(s) of fundamental quantities?

- I. Mass and weight
- II. Mass and length
- III. Time and current

- (A) I only
- (B) II only
- (C) I and III only
- (D) II and III only

3. The volume of a rectangular block of material which has dimension 400 cm x 400 cm x 125 cm is

- (A) 0.020 m^3
- (B) 20 m^3
- (C) 20 000 m^3
- (D) 20 000 000 m^3

4. The instrument MOST suitable for measuring the external diameter of a test tube is a

- (A) micrometer screw gauge
- (B) vernier caliper
- (C) metre rule
- (D) set square

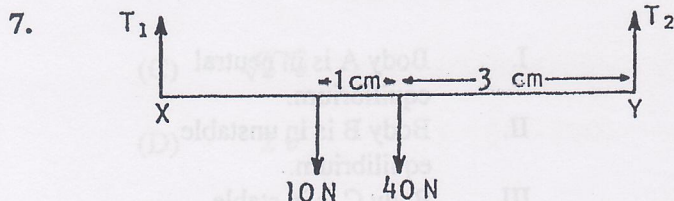
5. The density of a substance P is 0.8 times the density of water. Which of the following statements is/are true?

- I. The relative density of P is 0.8 $kg m^{-3}$.
- II. P will float in water.
- III. If a given volume of water is of mass 5 kg, an equal volume of P will have a mass of 4 kg.

- (A) I only
- (B) I and II only
- (C) II and III only
- (D) I, II and III

6. Which of the following is dimensionless?

- (A) Velocity
- (B) Relative density
- (C) Energy
- (D) Momentum



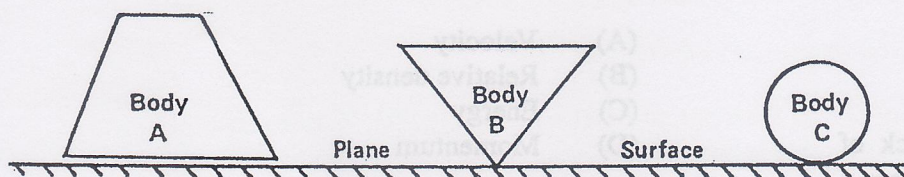
A uniform rod XY of weight 40 N is held in equilibrium, horizontally, as shown in the diagram above. Which of the following is true?

- (A) $T_1 < T_2$
- (B) $T_1 \leq T_2$
- (C) $T_1 = T_2$
- (D) $T_1 > T_2$

8. In which of the following is the quantity correctly paired with its unit?

	Quantity	Unit
(A)	Mass	Newton
(B)	Moment of force	Newton per metre
(C)	Energy	Newton-metre
(D)	Work done	Newton per metre

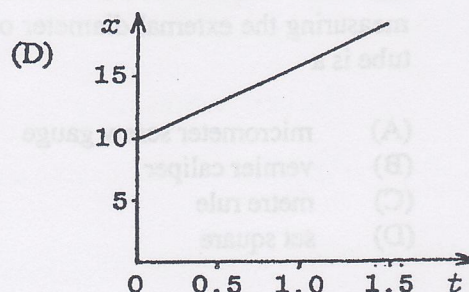
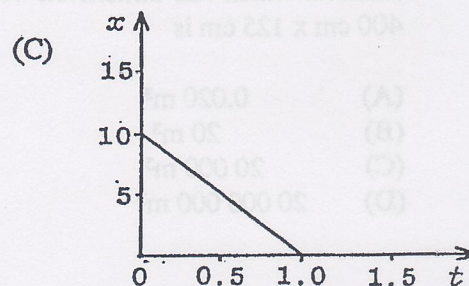
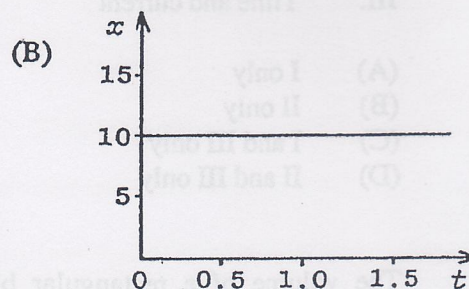
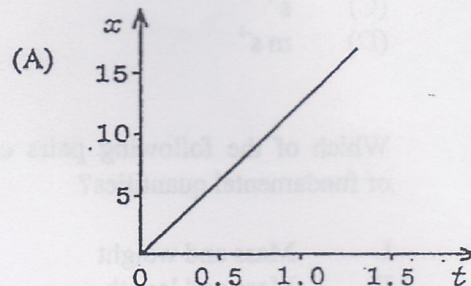
9. In relation to the diagrams below, which of the following statements is/are correct?



- I. Body A is in neutral equilibrium.
 II. Body B is in unstable equilibrium.
 III. Body C is in stable equilibrium.

- (A) II only
 (B) I and II only
 (C) II and III only
 (D) I, II and III

10. A vehicle is moving with a uniform velocity of 10ms^{-1} . This may be represented by which of the following graphs?



11. An object is thrown vertically upwards. At the highest point, which of the following is/are true?

I. Its velocity is zero.
 II. It is accelerating downwards
 III. No net force is acting

- (A) I only
 (B) I and II only
 (C) I and III only
 (D) I, II and III

12. The mass of a rifle is 100 times the mass of its bullet. If the recoil speed of the rifle is v , then the speed of the bullet is

- (A) $100 v^2$
 (B) $\frac{v^2}{100}$
 (C) $100 v$
 (D) $\frac{v}{100}$

13. Which of the following is an example of Newton's first law of motion?

I. An object falling freely in a vacuum
 II. An object moving at a constant speed in a circle
 III. An object being pulled along a rough surface with a constant velocity

- (A) II only
 (B) III only
 (C) I and II only
 (D) I and III only

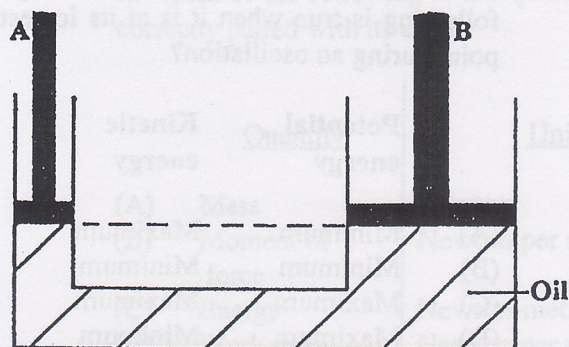
14. For a swinging pendulum, which of the following is true when it is at its lowest point during an oscillation?

	Potential energy	Kinetic energy
(A)	Minimum	Maximum
(B)	Minimum	Minimum
(C)	Maximum	Maximum
(D)	Maximum	Minimum

15. A body of mass m , moves with a velocity v , and has a certain kinetic energy. With what velocity would a body of twice this mass have to move in order to gain this same kinetic energy?

- (A) $\frac{v}{\sqrt{2}}$
 (B) $\frac{v}{2}$
 (C) $\sqrt{2} v$
 (D) $2 v$

16.



A and B in the diagram above, are two pistons which fit closely into the cylinders shown. Piston B has a cross-sectional area ten times that of piston A. Piston A is used to apply an excess pressure, $\mathcal{P}$, on the liquid under it. The excess pressure of the bottom of Piston B is

- (A) $\frac{\mathcal{P}}{10}$
- (B) $\mathcal{P}$
- (C) $10 \mathcal{P}$
- (D) $\frac{10}{\mathcal{P}}$

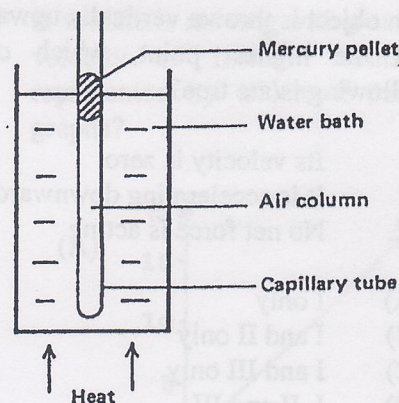
17. Item 17 refers to the following options which represent four units used to measure different quantities.

- (A) The newton (N)
- (B) The watt (W)
- (C) The joule (J)
- (D) The pascal (Pc)

EACH of these may be used more than once, once or not at all to answer the items below.

Which of the units is used to measure kinetic energy?

18.



The diagram above shows an air column trapped in a capillary tube by a mercury pellet. When the temperature of the water bath is raised, what happens to the volume and pressure of the trapped air?

- (A) Volume increases, pressure remains constant.
- (B) Volume increases, pressure increases.
- (C) Volume remains constant, pressure remains constant.
- (D) Volume remains constant, pressure increases

Item 19 refers to the following options.

- (A) Infra-red rays
- (B) Light rays
- (C) Sound waves
- (D) Water waves

In answering Item 19 a particular choice may be made more than once, once, or not at all. Which of the above

19. would have the shortest wavelength?

20. A student measured the temperature of a solution using two thermometers, one calibrated using a Celsius scale and the other a Kelvin scale. If the solution is heated and the temperature rises 5 degrees on the Celsius thermometer, the Kelvin thermometer reading will show a rise of

(A) 1 degree
(B) 5 degrees
(C) 278 degrees
(D) 268 degrees

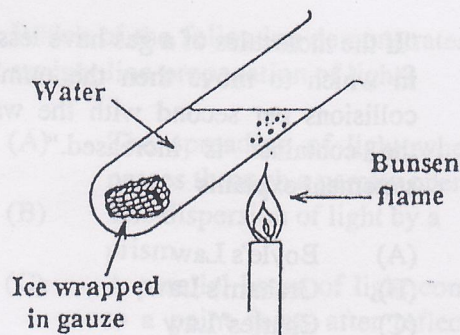
21. The heat capacity of a body depends on

(A) its mass and the specific heat capacity of the material
(B) its mass and the temperature change the body undergoes
(C) the density and temperature of the material
(D) the density and expansivity of the material

22. The unit(s) for specific latent heat of fusion would be

(A) J
(B) J kg^{-1}
(C) J K^{-1}
(D) $\text{J kg}^{-1} \text{K}^{-1}$

23.



The diagram above shows water boiling at the top of a glass test tube while a piece of ice remains unmelted at the bottom. Which of the following is a possible reason for this occurrence?

(A) Gauze is a poor conductor of heat.
(B) Water is a poor conductor of heat.
(C) water is a good conductor of heat.
(D) Glass is a good conductor of heat.

24. Which of the following occurs when one end of a metal rod is heated?

I. Hot molecule move down the rod.
II. Electrons move down the rod carrying energy.
III. Thermal energy passes from one molecule to the next.

(A) III only
(B) I and III only
(C) II and III only
(D) I, II and III

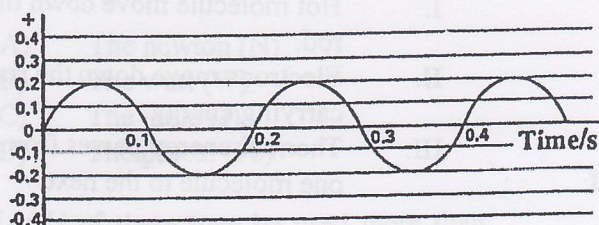
25. "If the molecules of a gas have less space in which to move then the number of collisions per second with the walls of the container is increased." This statement explains
- (A) Boyle's Law
 - (B) Graham's Law
 - (C) Charles' Law
 - (D) The Pressure Law

26. Which of the following is NOT implied by kinetic theory?

- (A) The internal energy of a gas depends only on the gas temperature.
- (B) The pressure of gas is caused by a change in momentum of the gas molecules bombarding the containing surfaces.
- (C) The kinetic energy is the same for all particles.
- (D) There are strong inter- molecular forces between two molecules close to each other.

Item 27 refers to the following displacement - time graph of a water wave.

Displacement/cm



27. What is the frequency of the wave?

- (A) 2.0 Hz
- (B) 2.5 Hz
- (C) 5.0 Hz
- (D) 10.0 Hz

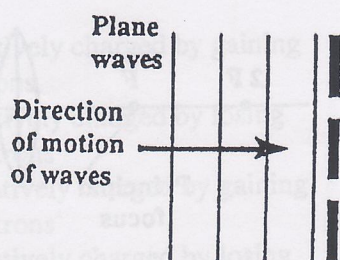
28. Which of the following statements about sound waves, water waves and electromagnetic waves is NOT correct?

- (A) They all transmit energy from the source to the surroundings.
- (B) They are all examples of transverse waves.
- (C) They all undergo reflection, refraction and diffraction.
- (D) They all have different speeds of propagation.

29. Which of the following would give the refractive index for light travelling from air to glass?

- I. $\frac{\text{Speed of light in glass}}{\text{Speed of light in air}}$
- II. $\frac{\text{Wavelength of light in air}}{\text{Wavelength of light in glass}}$
- III. $\frac{\text{Frequency of light in air}}{\text{Frequency of light in glass}}$
- (A) II only
- (B) I and II only
- (C) II and III only
- (D) I, II and III

30. A ripple tank is set up as shown in the diagram below.



Which of the following phenomena will be observed?

- I. Reflection
 - II. Diffraction
 - III. Interference
- (A) III only
 (B) I and II only
 (C) II and III only
 (D) I, II and III
31. Which of the following are NOT transverse waves?
- (A) Water waves on the sea
 (B) Light waves from a street lamp
 (C) Sound waves from a guitar string
 (D) Radio waves travelling from a Satellite
32. Infra-red radiation and sound waves are similar in that they
- (A) are both longitudinal waves
 (B) are both transverse waves
 (C) both transfer energy
 (D) are both produced by mechanical vibrations

33. Which of the following demonstrates the straight line propagation of light?

- (A) The spreading of light when it passes through a narrow opening
 (B) The dispersion of light by a prism
 (C) A parallel beam of light coming to a point focus after reflection from a spherical mirror
 (D) The operation of the pin-hole camera

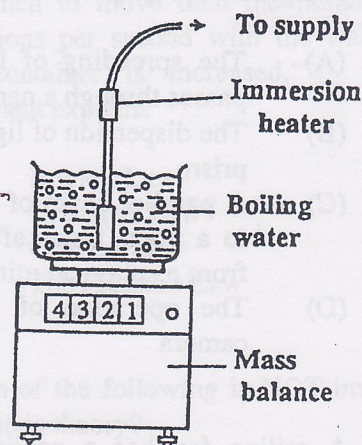
34. A ceiling fan has a cooling effect on people because

- (A) the temperature of the air is lowered
 (B) hot air rises
 (C) condensation can take place more readily
 (D) evaporation can take place more readily

35. The heat required to raise the temperature of a quantity of paraffin by 1 K is called its

- (A) specific heat capacity
 (B) heat capacity
 (C) specific latent heat
 (D) latent heat

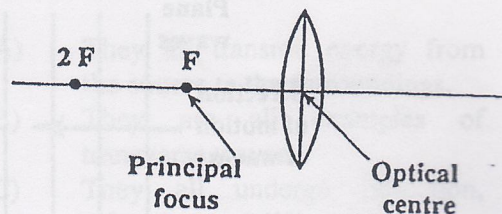
36.



Some water in a container (as shown) is brought up to its boiling point by an immersion heater. A watch is then started and the time, t , to boil off a mass, m , of water is found. If the heater has a power, P , the specific latent heat of vaporisation is

- (A) $\frac{Pt}{m}$
- (B) $\frac{P}{tm}$
- (C) $\frac{m}{Pt}$
- (D) Pmt

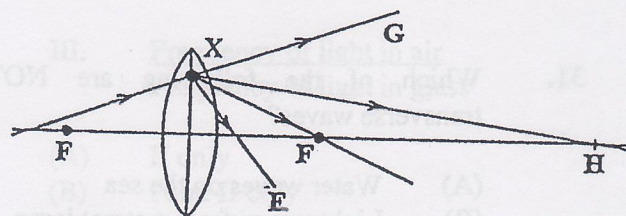
37.



From the diagram above, a real image is produced with a converging lens when the object is located

- (A) between the optical centre and F
- (B) between F and infinity
- (C) at F only
- (D) at 2F only

38.



In the diagram above, F represents the principal focus of a lens. Which line represents the ray of light after passing through the lens?

- (A) XE
- (B) XF
- (C) XG
- (D) XH

39. When a polythene rod is rubbed with a cloth, it becomes

- (A) positively charged by gaining protons
- (B) positively charged by losing electrons
- (C) negatively charged by gaining electrons
- (D) negatively charged by losing protons

40. A transformer is used to

- (A) convert alternating current to direct current
- (B) convert direct current to alternating current
- (C) alter the voltage
- (D) alter the frequency

41. Which of the following is true of the S.I. unit of electric current?

- I. It is called the coulomb.
- II. It is defined in terms of the force between two current-carrying wires.
- III. It is named after a scientist.

- (A) I only
- (B) I and II only
- (C) II and III only
- (D) I, II and III

42. An electronic flash which discharges in one millisecond passes 0.05 C of charge in this time. What is the average current that flows?

- (A) 500 A
- (B) 50 A
- (C) 5×10^{-1}
- (D) 5×10^{-5}

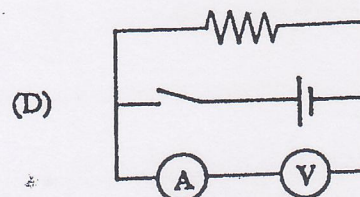
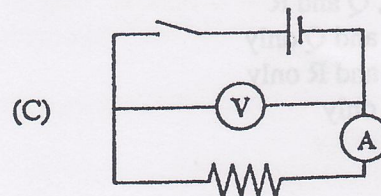
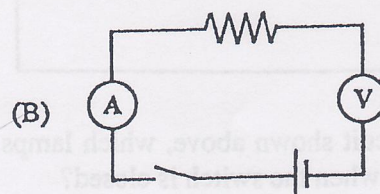
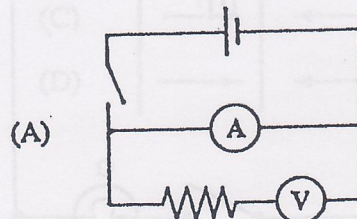
43. A beam of light passes from one medium to another of greater refractive index. When it refracts the

- (A) frequency changes
- (B) wavelength changes
- (C) speed increases
- (D) beam bends away from the normal

44. The critical angle for a material of refractive index 2 is

- (A) 90°
- (B) 42°
- (C) 30°
- (D) 0.5°

45. A student requires a circuit to measure the resistance of a resistor. Which of the circuits below is correctly connected?



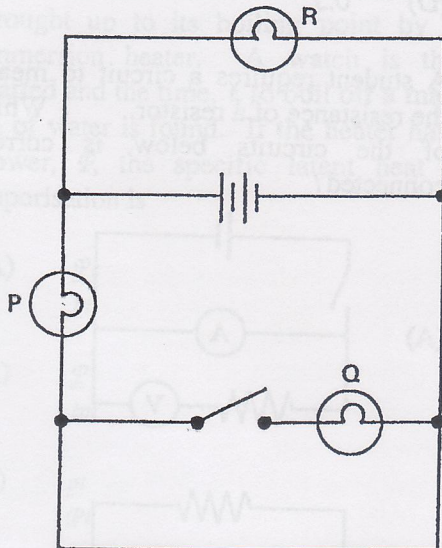
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46. Which of the following statements is/are true concerning a domestic wiring system?

- I. Switches are always connected in the live wire.
- II. Fuses are included to prevent short circuiting.
- III. The lights are connected in parallel with one another.

- (A) I only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

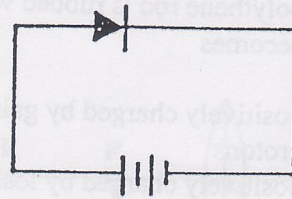
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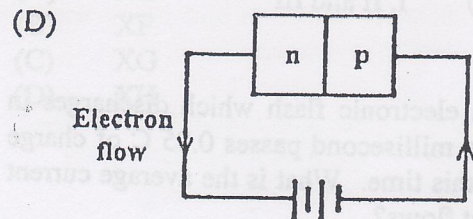
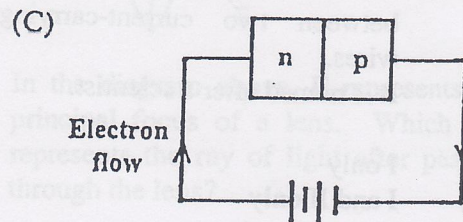
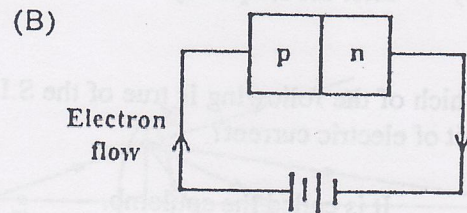
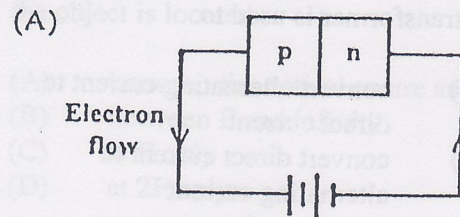
In the circuit shown above, which lamps will be lit when the switch is closed?

- (A) P, Q and R
- (B) P and Q only
- (C) P and R only
- (D) R only

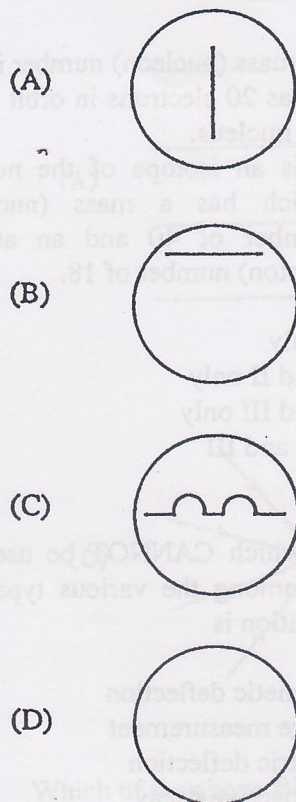
48.



The p-n junction is used as a semi-conductor diode. Which of the circuits below is the equivalent of the diagram above and also shows the correct direction of electron flow?



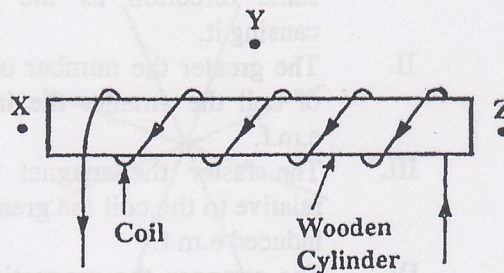
49. The following diagrams represent possible traces on the screen of a cathode-ray oscilloscope.



Which of the diagrams BEST represents an alternating voltage applied to the y-plates, with the time base off?

50. For which of the following would soft iron NOT be used?
- (A) A magnetic keeper
(B) The core of an electromagnet
(C) A magnetic shield
(D) A compass needle

Item 51 refers to the following diagram which shows a coil carrying a current and wrapped around a wooden cylinder.



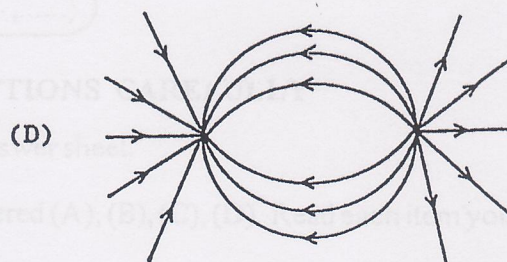
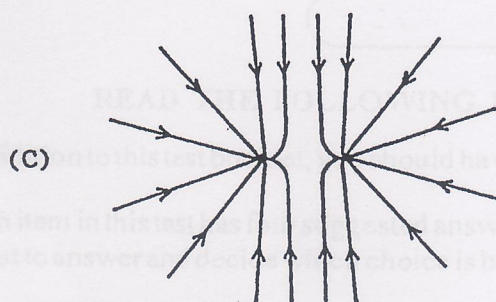
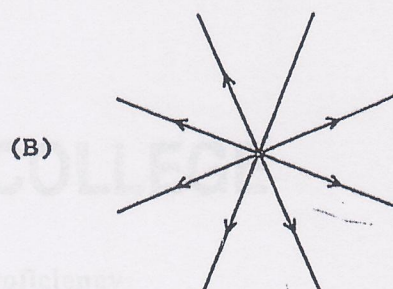
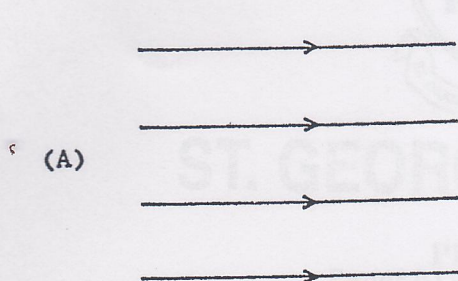
51. Which row of the table below shows the magnetic field directions at X, Y and Z?

	X	Y	Z
(A)	←	→	←
(B)	←	←	→
(C)	→	←	←
(D)	→	←	→

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52. Which of the following is true when a magnet is moved relative to a coil?
- I. The induced current is in the same direction as the charge causing it.
 - II. The greater the number of turns of coil the smaller the induced e.m.f.
 - III. The faster the magnet moves relative to the coil the greater the induced e.m.f.
 - IV. The stronger the magnetic field the greater the induced e.m.f.
- (A) I and IV only
(B) I and III only
(C) II and III only
(D) III and IV only
53. A magnetic field CANNOT be used to deflect
- (A) protons
(B) neutrons
(C) α -particles
(D) β -particles
54. Which of the following would induce an e.m.f. in a conductor?
- I. Moving a magnetic field across the conductor
 - II. Moving the conductor in the magnetic field of a magnetic
 - III. Placing the conductor near to a wire carrying an alternating current
- (A) I only
(B) I and II only
(C) I and III Only
(D) I, II and III
55. A neutral atom of Argon contains 18 protons and 20 neutrons. Which of the following statements concerning the atom is/are true?
- I. Its mass (nucleon) number is 38.
 - II. It has 20 electrons in orbit about the nucleus.
 - III. It is an isotope of the nuclide which has a mass (nucleon) number of 40 and an atomic (proton) number of 18.
- (A) I only
(B) I and II only
(C) I and III only
(D) I, II and III
56. A process which CANNOT be used to distinguish among the various types of nuclear radiation is
- (A) magnetic deflection
(B) range measurement
(C) electric deflection
(D) radioactive decay
57. A magnetic field can be used to deflect a path of
- (A) β -rays
(B) γ -rays
(C) X-rays
(D) light rays

Item 58 refers to the following options EACH of which may be used more than once, once or not at all. To answer the questions below.



Which of the above BEST represents

58. a uniform electric field?

59. Which of the following statements may be used to define the term "half-life"?

- I. It is half the time required by a nucleus to decay.
- II. It is the time for half of the nuclei of a sample to decay.
- III. It is the time for the activity of any random process to fall to one half of its original value.

- (A) I only
- (B) I and II only
- (C) II and III only
- (D) I, II and III

60. Which of the following scientists discovered the relationship $E = mc^2$?

- (A) Marie Curie
- (B) Isaac Newton
- (C) Albert Einstein
- (D) J. J. Thompson

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.